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# An Algorithm to Provide Systolic Bounds from Group Actions

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## Abstract

Conjectures state that Riemann surfaces with long systoles and big kissing number possess large holomorphic automorphism groups. Therefore, identifying and measuring geodesics explicitly is essential to fully understand the complex structure of a surface. In this work, we developed and implemented a computational routine that, starting from a group action, provides systolic bounds.

## Problem and strategy

The so-called Bolza surface $B$ is a special genus two surface: it possesses the largest automorphism group (of order 48), it has the longest systoles, and it has the largest number of systoles (12). Only for the Bolza surface these quantities are known explicitly; in general, even computing bounds is a difficult task.
The natural candidates to maximize the length and number of systoles are the Hurwitz's surfaces, which possesses the largest automorphism groups within their genus. In [1], Behn el al. developed a computational routine to construct a geometric representation for a surface starting from a group action. We re-implemented, documented and extended the algorithm to address the systolic bounding problem on concrete surfaces.

## Preliminaries

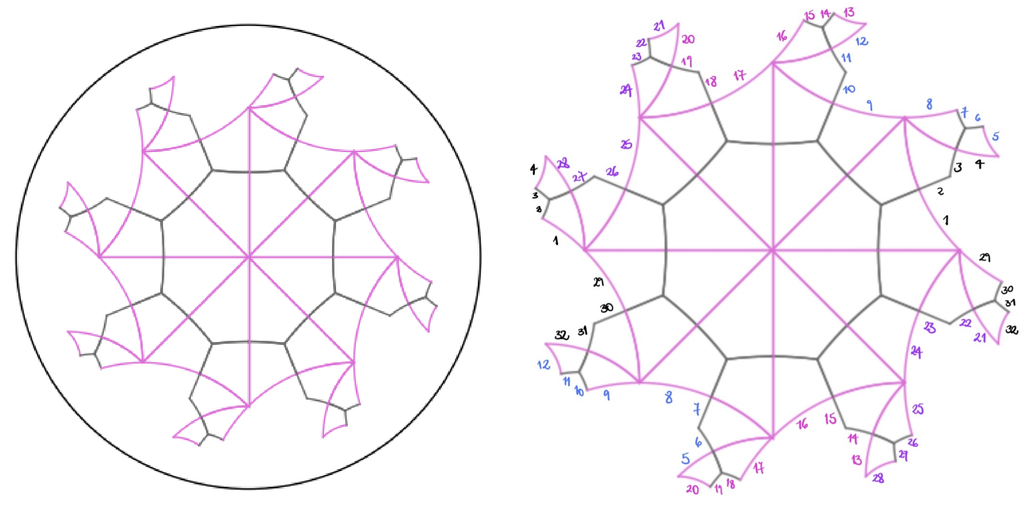
Let $S$ be a compact Riemann surface of genus at least two. By the Uniformization theorem, $S$ can be realized as a quotient of $\mathbb{D}$ by a Fuchsian group $\Gamma$.
A **systole** is a non-homotopically trivial simple closed geodesic of minimal length. The length of any systole is denoted by $\mathrm{sys}(S)$, and their number, up to free homotopy, by $\mathrm{Kiss}(S)$.
The hyperbolic metric $ds = \frac{2|dz|}{(1-|z|^2)}$ on $\mathbb{D}$ induces a metric on $S$ in such a way that $\mathrm{Aut}(S)$ acts on $S$ by isometries. Hence, the $\mathrm{Aut}(S)$-orbit of a systole consists of systoles.
In [4], the genus two surfaces with non-trivial group actions are classified. In the Bolza surface case, $\mathrm{Aut}(B) \cong \mathrm{GL}_2(3)$ acts with signature $(0; 8, 3, 2)$, equivalently, there is a map $\theta\colon \Delta \twoheadrightarrow \mathrm{GL}_2(3)$ from a Fuchsian group $\Delta$ that uniformizes the quotient orbifold $B/\mathrm{GL}_2(3)$. That $\Delta$ has presentation $\langle X, Y, Z \mid X^8 = Y^3 = Z^2 = XYZ = 1\rangle$.

## Algorithm's steps

- **Input:** Finite group $G$ and a signature for its action.
- **Output:** Fundamental domain for $\Gamma$ and a homology basis formed by systoles.

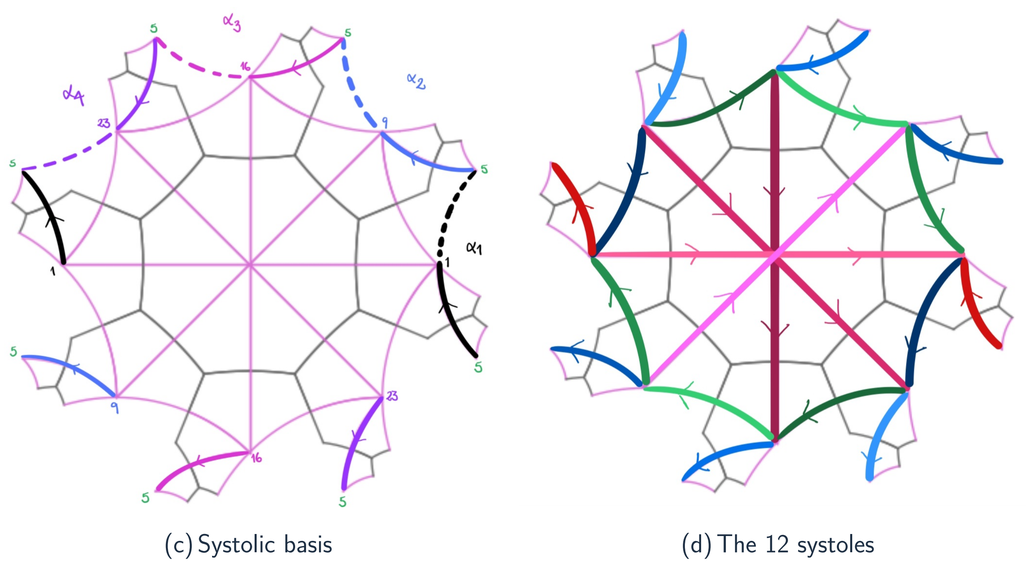
1. **Build a polygon**
   Realize $\Delta$ as a subgroup of Möbius transformations, generated by three rotations $X$, $Y$, and $Z$ of orders 8, 3 and 2. Define a surface-kernel epimorphism $\theta$ by selecting suitable images $\theta(X), \theta(Y), \theta(Z) \in \mathrm{GL}_2(3)$ (performed using GAP computational system). Draw a fundamental domain for $\Gamma = \ker(\theta)$.
2. **Side-pairing**
   Identifications of boundary vertices and edges according to $\Gamma$ (done-by-hand, with GAP assistance).
3. **Homology basis**
   Compute a basis for $\mathrm{H}_1(B)$ formed by boundary curves. Then, identify geodesic representatives on their homology classes (by hand).
4. **Systolic bounds**
   The length of any basis'element, which is a systole, bounds $\mathrm{sys}(B)$ from above, while the size of its $\mathrm{GL}_2(3)$-orbit bounds $\mathrm{Kiss}(B)$ from below.

## Algorithm in Pictures



(a) Fundamental domain

(b) Side-pairing



(c) Systolic basis

(d) The 12 systoles

## Results and Conclusions

The algorithm recovers the known bounds for Bolza surface, and along its theoretical foundations it provides a **strong computational tool** for understanding compact Riemann surfaces. Although our approach is promising for applications on higher genus surfaces, further research is needed to fully extend it. Future work might be focused on **automating done-by-hand steps** (side-pairing, computations in homology, geodesic identification), and on **adapting geometric constructions** to different signatures.

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